

Parameter Estimation Error: parameter estimation error is the standard error component from our non-linear regression models. As shown in Chapter 5, Estimating I-Star Model Parameters, there is some degree of uncertainty surrounding the parameters which will affect market impact estimates. For simplicity, we define the timing risk measure to only include the price volatility term when quoting the standard error of the market impact estimate but analysts conducting advanced sensitivity analysis may want to incorporate these additional components into the timing risk estimate. We have found the easiest way to determine the overall uncertainty is via Monte-Carlo simulation where volumes, intraday profile, price movement, and parameter values are sampled from historical observations and their estimated distribution. Investors performing this type of analysis may find that corresponding market impact uncertainty is much larger than simply the standard deviation of price movement.

TRADING COST MODELS—REFORMULATED

Market Impact Expression

Our market impact equations can be restated in terms of our trading strategies as follows:

I-Star

The I-Star calculation written in basis point and total dollar units is:

$$I_{bp}^* = a_1 \cdot \left(\frac{Q}{ADV} \right)^{a_2} \cdot \sigma^{a_3} \quad (7.10)$$

$$I_{\$/Share}^* = a_1 \cdot \left(\frac{Q}{ADV} \right)^{a_2} \cdot \sigma^{a_3} \cdot 10^{-4} \cdot P_0 \quad (7.11)$$

$$I_{\$}^* = a_1 \cdot \left(\frac{Q}{ADV} \right)^{a_2} \cdot \sigma^{a_3} \cdot 10^{-4} \cdot X \cdot P_0 \quad (7.12)$$

Market Impact for a Single Stock Order

The units of the market impact cost will be the same units as the instantaneous cost I-Star. Market impact cost for the three different trading strategy definitions is:

$$MI(POV) = \hat{b}_1 \cdot I^* \cdot POV^{\hat{a}_4} + (1 - \hat{b}_1) \cdot I^* \quad (7.13)$$

$$MI(\alpha) = \hat{b}_1 \cdot I^* \cdot \alpha^{\hat{a}_4} + (1 - \hat{b}_1) \cdot I^* \quad (7.14)$$